The influence of speaker goals on the interpretation of negation in gradable adjectives

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Models of language evolution characterize linguistic systems as shaped by multiple selection pressures arising from communicative use. While much work emphasizes efficient information transfer (e.g., Frank & Goodman, 2016; Carr et al., 2020), communicative use also involves social and strategic dimensions, in which agents may have diverse motives beyond informativeness (Mercier & Sperber, 2017). Argumentative pressures in particular have been modeled and empirically tested as shaping language use, primarily through analyses of how speakers design utterances strategically to influence listeners' beliefs or actions (e.g., Franke et al., 2012; Sumers et al., 2024; Macuch Silva et al., 2024). We extend this perspective by examining the listener's side. Specifically, we investigate whether listeners adjust their interpretation of (negated) gradable adjectives (e.g., *tall, not tall*, *not large*) based on the speaker's communicative goal. Gradable adjectives show substantial variation in their interpretation (Kennedy, 2007), which is further amplified under negation (Tessler & Franke, 2019). Second, we test whether the asymmetric interpretations of positive and antonymous gradable predicates in the scope of negation reported in the literature (e.g., *not tall* is often interpreted as *short*, whereas *not short* differs in meaning from *tall*) (Gotzner & Mazzarella, 2021; Alexandropoulou & Gotzner, 2024) partly reflect pragmatic inferences about speaker goals.

We conducted an online behavioral experiment (N = 200) examining whether interpretations of negated gradable adjectives are modulated by the speaker's communicative motive. Participants read short scenarios featuring a speaker describing either a person (*tall/short*, *playful/serious*) or an apartment

(*large/small*, *quiet/loud*). Each adjective pair appeared in four expression types: positive (*tall*), antonym (*short*), negated positive (*not tall*), and negated antonym (*not short*).

Scenarios implied one of three speaker goals: *amplifying* (benefiting from high degrees of property), *understating* (benefiting from low degrees of the property), or *neutral*. Participants then rated the described entity on a bipolar scale (e.g., extremely *short* → extremely *tall*). Each participant completed 12 trials combining adjective pair and goal, with one expression type per combination. The aim was to observe whether the speaker's goal influenced how participants interpreted negated constructions with gradable adjectives.

A linear regression predicted ratings by expression type, speaker goal, and their interaction. Results showed a significant effect of speaker goal: ratings were higher for understating speakers ($\beta = 0.04$, $t = 5.4$, $p < .001$) and lower for amplifying speakers ($\beta = -0.03$, $t = -4.15$, $p < .001$) relative to neutral. Thus, listeners inferred that an understating speaker's "not tall" likely referred to a taller individual than a neutral speaker's "not tall," while an amplifying speaker's "tall" was interpreted as less extreme. However, we found no support for the hypothesis that the semantic asymmetry between positive and negative adjectives is driven by speaker's goals (no significant interaction effects; $p > 0.25$).

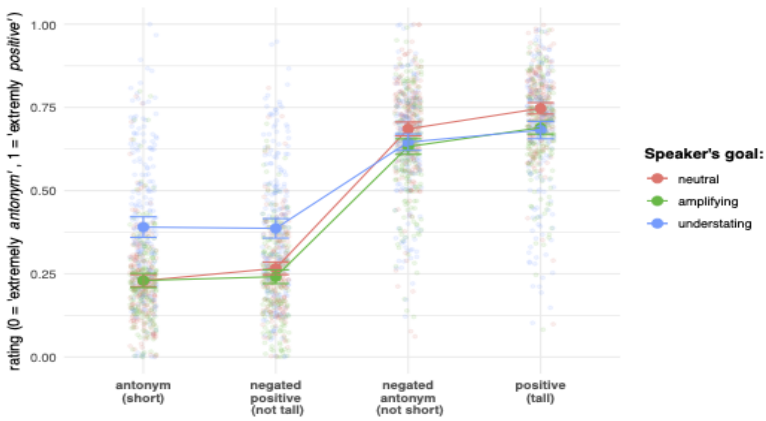

Figure 1. Average ratings of the object's property for each speaker (color) for the four expression types (x-axis). Error bars represent standard error and background dots show distribution of participants' raw data.

These findings indicate that listeners integrate speaker motives when interpreting evaluative language, such that amplifying and understating goals shift perceived truthfulness and interpretation toward or away from extremes. We find no

evidence that polarity asymmetries arise from pragmatic inferences about speaker goals. Our results therefore highlight the role of listener-side inferences in the communicative pressures shaping language use.

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